Case Study 3: ATM Transaction Sequence Simulation

1. Problem statement

Develop a Python program to simulate ATM transactions. The program should accept the initial bank balance and the number of transactions from the user. It should allow the user to deposit or withdraw money, check whether sufficient balance is available, update the account balance, and display the final balance.

2. Objectives

- To accept transaction details from the user.
- To perform deposit and withdrawal operations.
- To prevent withdrawals exceeding the available balance.
- To update the account balance after each transaction.
- To understand variables, input statements, arithmetic operators, conditional statements, and loops.

3. Assumptions

- The initial account balance cannot be negative.
- The number of transactions must be zero or greater.
- The user enters capital D for deposit and capital W for withdrawal.
- The transaction amount must be positive.
- Withdrawals exceeding the available balance are rejected.
- No transaction fees are included.

4. Algorithm

1. Start the program.
2. Accept the initial account balance.
3. Accept the number of transactions.
4. Check whether the balance and number of transactions are valid.

5. If the inputs are invalid, display an error message.
6. Otherwise, repeat the following steps for each transaction:
    - Accept the transaction type.
    - Accept the transaction amount.
    - If the amount is zero or negative, display an invalid amount message.
    - If the transaction type is D, add the amount to the balance.
    - If the transaction type is W, check whether sufficient balance is available.
    - If sufficient balance is available, subtract the amount from the balance.
    - Otherwise, display an insufficient balance message.
    - If the transaction type is invalid, display an error message.
    - Display the current balance.
7. Display the final balance.
8. Stop the program.

5. Flowchart



6. Pseudocode

START

INPUT balance

INPUT number_of_transactions

```
IF balance < 0 OR number_of_transactions < 0 THEN
  DISPLAY "Invalid input!"

ELSE
  FOR each transaction

    INPUT transaction_type
    INPUT amount

    IF amount <= 0 THEN
      DISPLAY "Invalid amount!"

    ELSE IF transaction_type = "D" THEN
      balance = balance + amount
      DISPLAY "Deposit successful."

    ELSE IF transaction_type = "W" THEN
      IF amount <= balance THEN
        balance = balance - amount
        DISPLAY "Withdrawal successful."
      ELSE
        DISPLAY "Insufficient balance!"
      END IF

    ELSE
```

```
        DISPLAY "Invalid transaction type!"
    END IF

    DISPLAY balance

  END FOR

  DISPLAY balance
END IF

STOP
```

7. Source code

```python
balance = float(input("Enter initial balance: Rs. "))
n = int(input("Enter number of transactions: "))

if balance < 0 or n < 0:
    print("Invalid input!")

else:
    for i in range(n):
        print("\nTransaction", i + 1)

        transaction_type = input( "Enter D for Deposit or W for Withdrawal: ")
        amount = float(input("Enter amount: Rs. "))

        if amount <= 0:
            print("Invalid amount!")

        elif transaction_type == "D":
            balance = balance + amount
            print("Deposit successful.")

        elif transaction_type == "W":
            if amount <= balance:
                balance = balance - amount
                print("Withdrawal successful.")
            else:
                print("Insufficient balance!")

        else:
            print("Invalid transaction type!")
        print("Current balance: Rs.", round(balance, 2))
    print("\nFinal balance: Rs.", round(balance, 2))
```

8.Test cases

Initial balance	Transaction	Expected result
₹1000	Deposit ₹500	Balance = ₹1,500
₹1,000	Withdraw ₹300	Balance = ₹700
₹500	Withdraw ₹800	Insufficient balance; balance = ₹500

9.Output

```
"C:\Users\tejas\PycharmProjects\Python Activity\.venv\Scripts\py
Enter initial balance: Rs. 1000
Enter number of transactions: 1

Transaction 1
Enter D for Deposit or W for Withdrawal: D
Enter amount: Rs. 500
Deposit successful.
Current balance: Rs. 1500.0

Final balance: Rs. 1500.0

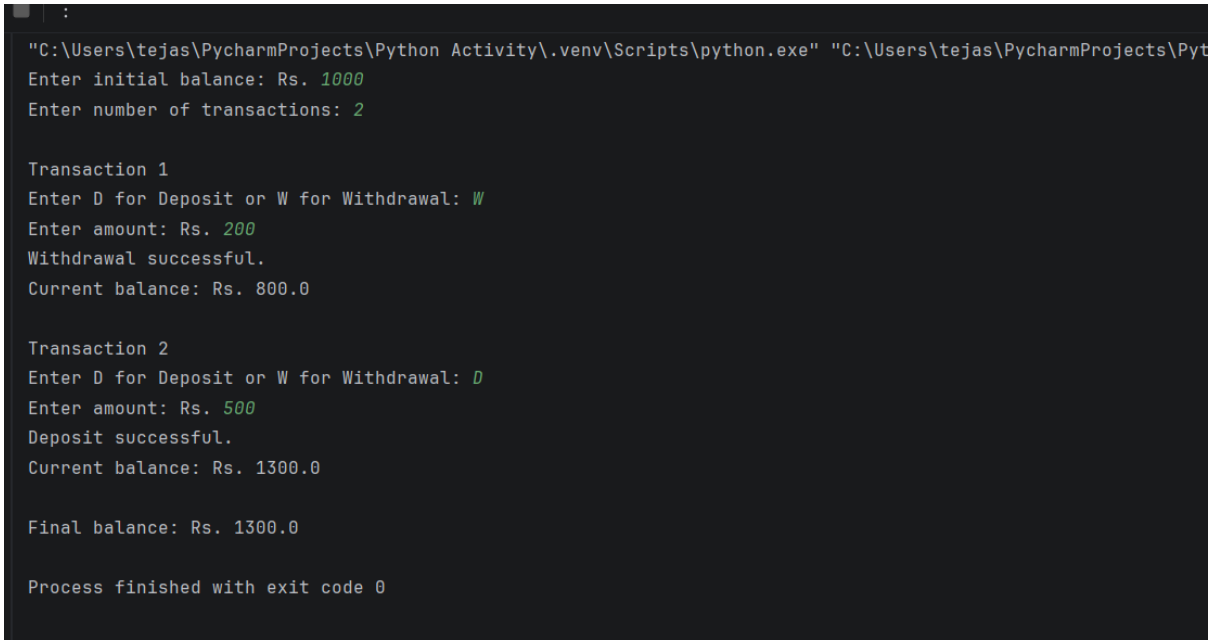
Process finished with exit code 0
```

```
Enter initial balance: Rs. 1000
Enter number of transactions: 1

Transaction 1
Enter D for Deposit or W for Withdrawal: W
Enter amount: Rs. 300
Withdrawal successful.
Current balance: Rs. 700.0

Final balance: Rs. 700.0

Process finished with exit code 0
```



10. Conclusion

The ATM Transaction Sequence Simulation program demonstrates basic banking operations such as depositing and withdrawing money. It updates the account balance after each transaction and prevents withdrawals when sufficient funds are unavailable. The program uses user input, variables, loops, conditional statements, and arithmetic operators to simulate a simple ATM system.